

10(a)	<i>any six from:</i> 1 small ; 2 water soluble so can move around cell ; 3 immediate source of energy ; A immediate energy donor 4 <i>ref. to</i> hydrolysed ; 5 phosphate removed releases energy ; 6 30.5 kJ mol^{-1} ; A molecule of ATP releases 30.5 kJ 7 ATP splits to ADP and P_i ; <i>note</i> $\text{ATP} \longrightarrow \text{ADP and } \text{P}_i + 30.5 \text{ kJ}$ gains mp6 and mp7 8 reversible ; 9 intermediate between, anabolic and catabolic reactions / energy yielding and energy requiring reactions ; 10 high turnover ;	6
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Question	Answer	Marks
10(b)	any nine from: 1 methylene blue / DCPIP, is a hydrogen acceptor (dye) ; 2 becomes colourless when reduced ; 3 use yeast suspension (in tube) ; 4 add named sugar (solution) and, methylene blue / DCPIP ; 5 put thin layer of oil on / put bung on, to prevent oxygen reaching yeast ; 6 <i>ref. to</i> water bath (at set temperature) ; 7 time how long it takes (for methylene blue / DCPIP) to go colourless ; 8 use colorimeter ; 9 <i>ref. to</i> 5 different temperatures ; 10 repeat (whole) experiment at least twice more ; 11 calculate mean values ; 12 method to calculate rate of respiration ; e.g. graph or $\frac{1}{T}$; 13 plot graph of rate of respiration against temperature ;	9